

Exhaust Gas Back Pressure Report

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Exhaust Gas Sources

Source	Mass Flow (kg/s)	Temp (°C)	Max BP (Pa)	Roughness
Main Engine	80.651	240	3000	Steel Welded
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System Summary (Combined)

Total Mass Flow	322.604 kg/s
Avg. Temperature	240 °C
System Max BP Limit	3000 Pa
Design Roughness	Steel Welded

Overall Assessment

Total Back Pressure	489.1 Pa
Total Back Pressure	49.875 mmWC
Compliance Status	PASSED
Safety Margin	83.7%

Pipeline Elements Breakdown

#	Element	Velocity (m/s)	ξ	ΔP (Pa)
1	Pipe	233.21	0.007	122.3
2	Pipe	233.21	0.007	122.3
3	Pipe	233.21	0.007	122.3
4	Pipe	233.21	0.007	122.3